



GNU nano 8.7.1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);
        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
akshitha_reddy@Akshitha-Reddy:~$ mkdir OS_LAB
akshitha_reddy@Akshitha-Reddy:~$ cd OS_LAB
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ nano task1.c
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ gcc task1.c -o task1
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.8 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [62.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [161 kB]
Fetched 574 kB in 4s (143 kB/s)
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 25
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 1688
Child Process
Child PID  : 1689
Child PID  : 1689
Executing command: ls
task1 task1.c
Child process completed.
```

```
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ uname -a
Linux Akshitha-Reddy 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               48 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      8
  On-line CPU(s) list:       0-7
Vendor ID:                   AuthenticAMD
Model name:                  AMD Ryzen 5 7235HS
CPU family:                  25
Model:                      68
Thread(s) per core:         2
Core(s) per socket:         4
Socket(s):                   1
Stepping:                    1
BogoMIPS:                    6387.81
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt
                             pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq
                             ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a mis
                             alignsse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed
                             adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushb
                             yasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm
Virtualization features:
  Virtualization:            AMD-V
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       128 KiB (4 instances)
  L1i:                       128 KiB (4 instances)
```

```
L2:                2 MiB (4 instances)
L3:                8 MiB (1 instance)
NUMA:
  NUMA node(s):    1
  NUMA node0 CPU(s): 0-7
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:         Not affected
  Indirect target selection: Not affected
  Itlb multihit:      Not affected
  L1tf:               Not affected
  Mds:                Not affected
  Meltdown:           Not affected
  Mmio stale data:    Not affected
  Old microcode:      Not affected
  Reg file data sampling: Not affected
  Retbleed:           Not affected
  Spec rstack overflow: Vulnerable: Safe RET, no microcode
  Spec store bypass:  Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:         Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:         Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSE-eIBRS Not affected; BHI Not affected
  Srbds:              Not affected
  Tsa:                Vulnerable: No microcode
  Tsx async abort:    Not affected
  Vmscape:            Not affected
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ ps
  PID TTY          TIME CMD
 1418 pts/0    00:00:00 bash
 1750 pts/0    00:00:00 ps
akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1         0  0 14:51 ?        00:00:00 /sbin/init
root           2         1  0 14:51 hvc0     00:00:00 /init
root           7         2  0 14:51 hvc0     00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          47         1  0 14:51 ?        00:00:00 /usr/lib/systemd/systemd-journald
systemd+     79         1  0 14:51 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root          83         1  0 14:51 ?        00:00:00 /usr/lib/systemd/systemd-udev
root         153         1  0 14:51 ?        00:00:00 snapfuse /var/lib/snapd/snaps/core22_2411.snap /snap/core22/2411 -o ro,nodev,allow_other,suid
root         154         1  0 14:51 ?        00:00:02 snapfuse /var/lib/snapd/snaps/snapd_27591.snap /snap/snapd/27591 -o ro,nodev,allow_other,suid
root         158         1  0 14:51 ?        00:00:00 snapfuse /var/lib/snapd/snaps/tree_54.snap /snap/tree/54 -o ro,nodev,allow_other,suid
```

```

root      184      1  0 14:51 ?      00:00:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root      185      1  0 14:51 ?      00:00:00 /usr/sbin/cron -f -P
message+  186      1  0 14:51 ?      00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root      190      1  0 14:51 ?      00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root      193      1  0 14:51 ?      00:00:00 /snap/snapd/current/usr/lib/snapd/snapd
root      194      1  0 14:51 ?      00:00:00 /usr/lib/systemd/systemd-logind
syslog    212      1  0 14:51 ?      00:00:00 /usr/sbin/rsyslogd -n -iNONE
root      229      1  0 14:51 tty1    00:00:00 /usr/sbin/agetty --noproset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/lib/issue.d
root      245      1  0 14:51 ?      00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
_chrony   250     184  0 14:51 ?      00:00:00 /usr/sbin/chronyd -n -F 1 -x
_chrony   263     250  0 14:51 ?      00:00:00 /usr/sbin/chronyd -n -F 1 -x
root      388      2  0 14:51 ?      00:00:00 login -- akshitha_reddy
akshith+  429      1  0 14:51 ?      00:00:00 /usr/lib/systemd/systemd --user
akshith+  431     429  0 14:51 ?      00:00:00 (sd-pam)
akshith+  459     388  0 14:51 pts/1    00:00:00 -bash
root     1305      1  0 15:22 ?      00:00:00 /usr/libexec/wsl-pro-service
root     1415      2  0 15:24 ?      00:00:00 /init
root     1416    1415  0 15:24 ?      00:00:00 /init
akshith+  1418    1416  0 15:24 pts/0    00:00:00 -bash
polkitd   1556      1  0 15:25 ?      00:00:00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
akshith+  1760    1418  0 15:31 pts/0    00:00:00 ps -ef

```

```

akshitha_reddy@Akshitha-Reddy:~/OS_LAB$ top
top - 15:32:02 up 40 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 30 total, 1 running, 29 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 5769.6 total, 5028.0 free, 539.4 used, 345.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 5230.2 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24560	15604	11544	S	0.0	0.3	0:00.80	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50416	16944	15640	S	0.0	0.3	0:00.24	systemd-journal
79	systemd+	20	0	22416	14440	11944	S	0.0	0.2	0:00.09	systemd-resolve
83	root	20	0	35600	12568	9252	S	0.0	0.2	0:00.77	systemd-udevd
153	root	20	0	159348	1864	1540	S	0.0	0.0	0:00.00	snappfuse

```

top - 15:33:18 up 41 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 30 total, 1 running, 29 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 5769.6 total, 5009.5 free, 557.9 used, 345.7 buff/cache

```

MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 5211.4 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24560	15604	11544	S	0.0	0.3	0:00.80	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50416	16944	15640	S	0.0	0.3	0:00.24	systemd-journal
79	systemd+	20	0	22416	14440	11944	S	0.0	0.2	0:00.09	systemd-resolve
83	root	20	0	35600	12568	9252	S	0.0	0.2	0:00.77	systemd-udev
153	root	20	0	159348	1864	1540	S	0.0	0.0	0:00.00	snappfuse
154	root	20	0	454144	12376	1516	S	0.0	0.2	0:02.10	snappfuse
158	root	20	0	159348	1880	1556	S	0.0	0.0	0:00.00	snappfuse
184	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.05	chronyd-starter
185	root	20	0	4472	3108	2860	S	0.0	0.1	0:00.01	cron
186	message+	20	0	8824	5376	4676	S	0.0	0.1	0:00.06	dbus-daemon
190	root	20	0	44096	29704	14556	S	0.0	0.5	0:00.25	networkd-dispat
193	root	20	0	2220748	38156	24808	S	0.0	0.6	0:00.82	snapp
194	root	20	0	18436	9472	8228	S	0.0	0.2	0:00.09	systemd-logind
212	syslog	20	0	220548	5996	4632	S	0.0	0.1	0:00.08	rsyslogd
229	root	20	0	5492	2748	2564	S	0.0	0.0	0:00.00	agetty
245	root	20	0	123460	32568	17956	S	0.0	0.6	0:00.22	unattended-upgr
250	_chrony	20	0	23732	11096	7100	S	0.0	0.2	0:00.08	chronyd
263	_chrony	20	0	12096	2432	1792	S	0.0	0.0	0:00.00	chronyd
388	root	20	0	8644	5556	4716	S	0.0	0.1	0:00.01	login
429	akshith+	20	0	22772	13464	10988	S	0.0	0.2	0:00.14	systemd
431	akshith+	20	0	22916	4096	2096	S	0.0	0.1	0:00.00	(sd-pam)
459	akshith+	20	0	6136	5412	3864	S	0.0	0.1	0:00.02	bash
1305	root	20	0	1857908	15260	12552	S	0.0	0.3	0:00.07	wsl-pro-service
1415	root	20	0	3188	1124	980	S	0.0	0.0	0:00.00	SessionLeader
1416	root	20	0	3204	1260	1108	S	0.0	0.0	0:00.03	Relay(1418)
1418	akshith+	20	0	6268	5540	3872	S	0.0	0.1	0:00.07	bash
1556	polkitd	20	0	306444	8440	7628	S	0.0	0.1	0:00.08	polkitd
1770	akshith+	20	0	8172	6008	3900	R	0.0	0.1	0:00.05	top